

Paper Review Form for *Landscape Research Record 15* (updated 5-10-2026)

The primary objective of this peer review process is to maintain the high quality and scholarly rigor of CELA's Landscape Research Record (LRR). Your thorough and insightful evaluation is essential in upholding these standards, ultimately contributing to a valuable and respected publication within the field.

- This review process adheres to a double-blind format. Therefore, even if you happen to identify the author(s) of the manuscript, you must maintain the confidentiality of all review-related information.
- Your feedback should be both constructive and comprehensive, offering clear guidance to the author(s) on how to improve their work. Please remember that a significant portion of submissions may come from early-career faculty and graduate students. Your thoughtful and encouraging critique can play a vital role in their professional development. If you believe the manuscript should be rejected, it is crucial to articulate the reasons for your recommendation.
- Kindly provide all your feedback to the author within the structured format of the designated evaluation form. For each section, please directly address the question posed and immediately follow with specific and actionable suggestions for improvement. While detailed line-by-line edits are not required, you may refer to specific content within the manuscript by indicating line numbers/page numbers.
- Furthermore, please provide constructive feedback that the author(s) can utilize to strengthen their future research and writing endeavors.
- If the subject matter of the paper falls outside your area of expertise, we ask that you approach the review as an educated reader from another field. In such cases, your primary focus should be on assessing and suggesting improvements to the clarity and accessibility of the writing for a broader academic audience.
- **Based on the newly passed CELA AI policy, reviewers are strictly prohibited from inputting any article, abstract, or manuscript into any AI platform to automatically generate reviews. Doing so constitutes a direct breach of author copyright and confidentiality. However, reviewers are permitted to use AI tools solely to edit, refine, or adjust the phrasing of their own original review writing.**
- We deeply appreciate your commitment to this process and the time you dedicate to reviewing. Please submit your evaluation by the indicated deadline. We would be grateful for your early submission of feedback.

Paper ID / Number: **392** (Available in the document file name)

Title of the paper reviewed: **STRUCTURING THE DIGITAL TWIN: USING SITES V2 AS A DATA ARCHITECTURE FOR CAMPUS LANDSCAPES**

Recommendation to track chair(s) and CELA Vice President for Research and Creative Scholarship:

- A. Accept _____
- B. Accept with revisions X
- C. Reject _____

- 1. ABSTRACT:** Does the abstract clearly and comprehensively describe the purpose of the study, methods used, key results, and main conclusions, effectively capturing the paper's essence? What specific revisions could the authors make to enhance its clarity, completeness, or conciseness?

I think the abstract is very complete and thorough. One change I would recommend is that there is an awkward sentence:

Your sentence: "Field reconnaissance further identified a subsurface steam corridor not detectable via remote sensing, underscoring the need for mixed-method ground-truthing even in data-rich workflows."

My recommended edits: "Field reconnaissance further identified a subsurface steam corridor undetectable by remote sensing, underscoring the need for mixed-method ground-truthing even in data-rich workflows."

- 2. BACKGROUND:** Does the background section provide sufficient and relevant context for the study? How could the authors improve the background to better establish the significance and relevance of their work?

Yes, the background section does a solid job of presenting the relevant work. However, I propose the following to enhance the background.

1. In section 1.1. The following is confusing and hard to digest, consider rewriting: "The SITES v2 rating system operationalizes this framework: when a student proposes a design intervention and the SITES scoring engine evaluates it against quantifiable ecological standards, the system reproduces the essential structure of Schön's reflective cycle in a computational context. Phase 1 constructs the data foundation—the structured representation of the situation—that makes this reflective exchange possible in Phase 2. The adoption of Public Participation GIS (PPGIS) principles (Ramirez Aranda et al., 2023) extends Schön's reflective conversation to include the situated knowledge of campus users, broadening both the data foundation and the range of feedback the system can provide." It comes across as fragmented and disconnected on the first few reads. Try to polish and lead transitions to a smoother reading experience and, by extension, a quicker understanding of your intent.
2. In 1.4.1: Update the following sentence to read more clearly: "Shahat et al. (2021) reviewed city digital twin potentials, identifying integration challenges that apply equally to site-scale landscape systems."

Suggested edit: "Shahat et al. (2021) reviewed the potential of city digital twins, identifying integration challenges that apply equally to site-scale landscape systems." This sentence should be followed by another sentence that more clearly articulates the shared challenges.

3. In 1.4.1: Change, "...twin depends less on novelty..." to "...twin depends less on the novelty..."
4. 1.4.1 could use a few more sentences to highlight past and existing landscape digital twin efforts, their benefits, and their pitfalls. Otherwise, this section is a bit too vague to explain why sites are a good organizing framework and why they support the case you build in 1.4.3. There are several articles in the Journal of Digital Landscape Architecture that can support this deeper review of previous modeling efforts and their pitfalls.

5. In 1.4.2, you state, “Weber (2024) demonstrates how mobile LiDAR and thermal imaging extend documentation capabilities.” And? What were these extended capabilities, and why are they important to use as designers in your digital twin effort? Please don’t assume your audience knows these reasons. You have a number of such statements in this section that need further elaboration. I don’t want your important arguments to get lost and dismissed by those who have had successful careers designing without this tech. The George and Park sentence is a good example of what you need to do with your other sentences, because it is followed by, “...finding that students develop enhanced spatial awareness and site analysis skills.”
6. In 1.4.2, you state, “The complementarity of photogrammetric platforms—visual fidelity from mesh-based processors and survey-grade accuracy from photogrammetric calibration with ground control points (GCP) integration—is essential for landscape digital twins that must serve both design exploration and spatial analysis (Colomina & Molina, 2014).” That this is essential, but never explain why. Please elaborate for those who are not digitally inclined.
7. In 1.4.3, you spend a lot of time elaborating on what Sites is, but not enough on justifying its benefits as a framework for digital twins and on how this integrates with the organization of your data and methods that follow. You need this elaboration because the counterargument will be that creating a digital twin is expensive and time-consuming. Can’t I just use the checklist?

3. PURPOSE/CONTRIBUTION: Are the purpose and objectives of this paper clearly and unambiguously stated? In your expert opinion, does the main contribution of this work have the potential to advance scholarship within the field of landscape architecture? Please suggest specific changes the authors could make to enhance the clarity of their purpose and objectives, and offer recommendations on how they might further emphasize the significance and potential impact of their contribution to the field.

Yes, these are very clearly articulated. I would only suggest that you add a sentence or two to in 1.3 to present what is the purpose and contribution beyond this specific project and grant to the wider field of landscape architecture: i.e., if a successful twin can be built using this approach designers can more rapidly understand possible outcomes of design interventions with a greater degree of certainty before construction even begins and how that might link to sustainability, biodiversity, etc.

4. REVIEW OF LITERATURE: Does the author adequately acknowledge and credit relevant prior work, and is the current study appropriately positioned within the existing body of knowledge? How could the authors enhance the literature review to provide a more comprehensive and insightful context for their contribution?

My comments on the literature review are in the background section of the comments above (Section 2. Background), as these sections were integrated by the authors.

5. METHODS: Are the research methods described with sufficient clarity and detail to allow for replication, and are the chosen methods appropriate for the study's purpose? What specific details or clarifications could the authors add to the methods section? Are there any concerns about the appropriateness of the chosen methods?

Overall the method structure is sound. I have a few suggestions to improve the clarity of your methods and their broader impact.

1. 2.1 – Try to add a few sentences that again broaden this site’s representation beyond your specific project. For example, something to the effect, “While this site and its context is unique to NC A&T many landscape architects and land managers face similar concerns on other multifunctional facilities at other campuses, schools, and community centers.”
2. 2.3 – Please try to provide a bit more about UAV flight parameters and other specific items, such as what soil characteristics were specifically integrated or collected.

6. PRESENTATION OF DATA (Tables, Figures, and Images): Are all tables, figures, and images clearly labeled, appropriately placed within the text, and effectively contribute to the reader's understanding of the data? What specific improvements could be made to their labeling, placement, or clarity?

1. The order of some of your figures and tables seems out of sequence. In the site introduction, it would be worth noting the different zones in the image. The reason is that you then have a series of tables that reference the different zones, but the plant with the zones (Fig. 3)

7. FINDINGS/CONCLUSIONS: Are the findings presented clearly and logically, and are the conclusions adequately supported by the data? How could the authors improve the presentation of their findings or the articulation of their conclusions? Are there any discrepancies between the data and the conclusions drawn?

I would like to see an expansion on the demonstrated benefits and on what was found through the digital twin, as well as what didn’t go well. This will help reinforce the impacts you mentioned in your abstract and the opening section of your article. Also, mention future steps for improvement, including methods, data collection, integration, and AVA’s usefulness itself.

8. OVERALL STRUCTURE/CLARITY: Is the overall structure of the paper logical and easy to follow, with content organized into appropriate and clearly labeled sections? Is the writing clear, concise, and free of unnecessary jargon? Does the title accurately reflect the paper's content, and does the content effectively present the study's results without over-reliance on lists or bullet points in the main text? Additionally, is the abstract within the specified word limit? Please provide specific suggestions on how aspects of the paper's structure, writing, and content organization could be improved for better clarity and flow, and whether the title accurately represents the study.

Overall, the structure and clarity are solid, aside from the comments noted above and Section 2.3. The structure of 2.3 is hard to follow because it presents a list of data without explaining how to interpret it or how it connects to the authors' decisions about data collection.

9. FORMATTING AND STYLE: Does the document adhere to the specified formatting guidelines for margins, font, line spacing, and indentation of paragraphs and titles/subtitles? Please identify any deviations from the required formatting style and suggest specific corrections. Do they adhere to any specified limits on the number of tables and figures? Please provide specific feedback on the placement, labeling, clarity, and

adherence to any limitations regarding the number of tables and figures. (See published LRR paper and template here: <https://thecela.org/landscape-research-record/>)

The formatting is good.

10. CITATIONS AND REFERENCES (APA Style) Do the in-text citations and the reference list consistently adhere to the specified APA style guidelines, including proper matching of citations and references, alphabetical ordering of the reference list, and the use of hanging indents? Please point out any inconsistencies or errors in the application of APA style for citations and references.

Seems good.

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Recommendation to track chair(s) and CELA Vice President for Research and Creative Scholarship:

- A. Accept _____
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1. ABSTRACT: Does the abstract clearly and comprehensively describe the purpose of the study, methods used, key results, and main conclusions, effectively capturing the paper's essence? What specific revisions could the authors make to enhance its clarity, completeness, or conciseness?

The abstract clearly and comprehensively describes the purpose of the study while also identifying the data sources and summarizing the findings. It provides a strong overview of the paper. One suggestion that I would offer would be that the abstract should more clearly explained the difference between data architecture proposed in the paper and the components that are under development within the larger context of the literature. For example the abstract states that six data streams were integrated through SITES V2 baselines but the paper doesn't completely present the complete site score for all the four sections. I'd recommend making this distinction stronger in the abstract by stating that this is a single-site case study which identifies four SITES sections (used for data collection).

2. BACKGROUND: Does the background section provide sufficient and relevant context for the study? How could the authors improve the background to better establish the significance and relevance of their work?

The background is well-structured and the research questions look solid and measurable within the study. There is a bit of a mismatch between the opening statement and what is covered in the background as well as the rest of the paper. It stated in the research objectives that one goal of the project is to develop the digital twin as a pedagogical tool for landscape architecture education however it does not seem that an evaluation of learning objectives or competencies learned by students was measured within this case study. I'd suggest either narrowing the objectives mentioned in the paper or include the measured outcomes of student learning/achievement/pedagogical outcomes.

I agree that Evans-Allen funding context is important for the paper but it could be shortened or move to the study context so that the background remains the focus to address the problem in the literature/scholarship. Also it looks like public participation GIS is introduced as a theoretical framework but it isn't utilized or explained much in the methods or results I would either remove it or show evidence of it directly in the methods and results.

3. PURPOSE/CONTRIBUTION: Are the purpose and objectives of this paper clearly and unambiguously stated? In your expert opinion, does the main contribution of this work have the potential to advance scholarship within the field of landscape architecture? Please suggest specific changes the authors could make to

enhance the clarity of their purpose and objectives, and offer recommendations on how they might further emphasize the significance and potential impact of their contribution to the field.

In order to strengthen the contribution of this paper and the work presented, I would suggest presenting the detail in which the study used with a workflow diagram or a data crosswalk which connects each SITES criterion to a dataset, variable(s), resolution, and/or a validation procedure. This would make the work more transferable for future researchers. It would also allow for SITES to include this as an additional metric.

- 4. REVIEW OF LITERATURE:** Does the author adequately acknowledge and credit relevant prior work, and is the current study appropriately positioned within the existing body of knowledge? How could the authors enhance the literature review to provide a more comprehensive and insightful context for their contribution?

The literature review acknowledges relevant work that's presented in components of the paper however it's currently brief and primarily descriptive. The literature review should synthesize existing approaches and mention the limitations with support the knowledge gap that you were addressing with this paper. Schon's reflective approach is relevant to student learning and design feedback, however it isn't clear that you will use this as a pedagogical assessment or method for content delivery in a future course or studio. Perhaps spelling this out would make the study more compelling.

- 5. METHODS:** Are the research methods described with sufficient clarity and detail to allow for replication, and are the chosen methods appropriate for the study's purpose? What specific details or clarifications could the authors add to the methods section? Are there any concerns about the appropriateness of the chosen methods?

The methods provide a great and appropriate manner for data collection, examining performance, and spatially explicitly explicit site conditions; however, the methods aren't explained in enough detail to provide replication of this study. It would be interesting and useful to see how the sequence unfolded for one site in phases: (e.g., Data acquisition, Processing, Integration, Validation, then dissemination). It would also be useful if the research team could provide details about field reconnaissance explaining the number of visits, observers, measurements collected and potentially the procedure of processing the data. Also, if available, a link to a data repository and metadata or even a data availability statement might help to explain the transparency of the study.

- 6. PRESENTATION OF DATA (Tables, Figures, and Images):** Are all tables, figures, and images clearly labeled, appropriately placed within the text, and effectively contribute to the reader's understanding of the data? What specific improvements could be made to their labeling, placement, or clarity?

For figures tables and images I would recommend a few visual improvements. Figures 2-3 and four occur several pages after the first mention in the results. Move them closer to the mention. The scale in Figure 2 is difficult to read. I would remove data from the map that isn't relevant to the site information presented in

the paper. The font in Figure 3's legend is too small. Text should be at 8 point and perhaps use call outs on the drawing. Figure 4 needs a north arrow, scale bar and an annotation denoting the steam vent corridor.

Also figure 5 is engaging and interesting as a graphic user interface for the proposed tool, however The text is too small to read and potentially the graphic might require a full page and landscape or a citation/ GIT Hub doi link. The alternative to this would be creating a diagram which explains the data architecture developed during the first phase of the project.

Revise table to so that it distinguishes between data that were acquired processed validated and integrated. For Table 3 please provide definitions for the condition categories coverage estimates and risk classifications.

7. FINDINGS/CONCLUSIONS: Are the findings presented clearly and logically, and are the conclusions adequately supported by the data? How could the authors improve the presentation of their findings or the articulation of their conclusions? Are there any discrepancies between the data and the conclusions drawn?

The findings were organized logically however here are some suggestions for improvement. The researchers mentioned the data architecture as being non redundant, however a redundancy analysis was not presented. Again, an explanation of the data architecture within a detailed explanation would provide a stronger evaluation of each SITES Criterion to support transferability of this study. Please revise the results illustrate the direct answer for each research question presented at the beginning of the paper. Separate observed findings from interpretation and please identify which conclusions are made from measured data versus professional observation versus AI assisted classification, etc. The conclusion should focus directly on what was covered during this specific study. Also the discussion is interesting but remains speculative. I would recommend proposing this as future work along with measured pedagogical outputs aligned with the objectives mentioned earlier on in the paper.

8. OVERALL STRUCTURE/CLARITY: Is the overall structure of the paper logical and easy to follow, with content organized into appropriate and clearly labeled sections? Is the writing clear, concise, and free of unnecessary jargon? Does the title accurately reflect the paper's content, and does the content effectively present the study's results without over-reliance on lists or bullet points in the main text? Additionally, is the abstract within the specified word limit? Please provide specific suggestions on how aspects of the paper's structure, writing, and content organization could be improved for better clarity and flow, and whether the title accurately represents the study.

The clarity within the paper is presented well however here are a few suggestions for next steps. I would recommend emphasizing the specific research gap that you are addressing in the paper. The Evans Allen discussion can be incorporated within the study context but out of the objectives/questions section. I'd recommend removing the conference feedback section.

There is quite a bit of repetition in the paper regarding SITES V2 as a data specification. I would state the argument once in the introduction demonstrate it through the methods and results and then illustrate the interpretation of the findings in the discussion.

9. FORMATTING AND STYLE: Does the document adhere to the specified formatting guidelines for margins, font, line spacing, and indentation of paragraphs and titles/subtitles? Please identify any deviations from the required formatting style and suggest specific corrections. Do they adhere to any specified limits on the number of tables and figures? Please provide specific feedback on the placement, labeling, clarity, and adherence to any limitations regarding the number of tables and figures. (See published LRR paper and template here: <https://thecela.org/landscape-research-record/>)

Please review the CELA LRR instructions and template. Adhere to specifics regarding tables figures formatting and legibility.

10. CITATIONS AND REFERENCES (APA Style) Do the in-text citations and the reference list consistently adhere to the specified APA style guidelines, including proper matching of citations and references, alphabetical ordering of the reference list, and the use of hanging indents? Please point out any inconsistencies or errors in the application of APA style for citations and references.

It looks as though there are several inconsistencies regarding APA citation styles and some of the references used. Please correct them before resubmitting. ALSO, please verify that each citation supports the claim associated with it. For example, Tallamy and the SITES manual, as stand alone citations, do not. Please provide additional supporting literature.